# MATERIALS
## OF THE XXI INTERNATIONAL SCIENTIFIC AND PRACTICAL CONFERENCE

# MODERN SCIENTIFIC POTENTIAL - 2024

February 28 - March 7 , 2024

**Volume 3**

SHEFFIELD
SCIENCE AND EDUCATION LTD
2024

**Pil E.A.**
*Academician RANH, Professor, Dr. Sc. Eng., St. Petersburg*

## HOW NEGATIVE VALUES OF VARIABLES AFFECT X4 CALCULATIONS AND THE PLOTTING OF 3D GRAPHS VEL

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. Vel (GDPel) = f(X1, X2, X3, X4, X5). In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows X1= X3= -1...-10, X2= -10...-1, X4= 2.78...0.76, X5=-0.1...-1. In this example, Vel is exaggerated by 27.29 times. In the following Figure 2 the depicted 3D Vel graph at variables X1= X3= -10...-1, X2= -1...-10, X4=0.29...2.87, X5= -0.1...-1 increases 1000 times.

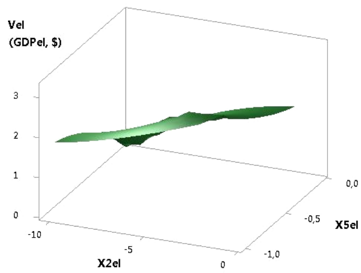


Fig. 1. 3D of Vel (GDPel)

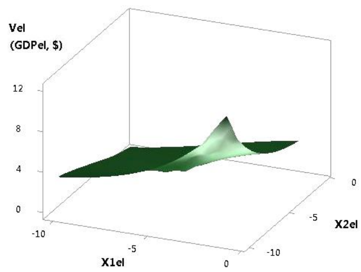


Fig. 2. 3D of Vel (GDPel)

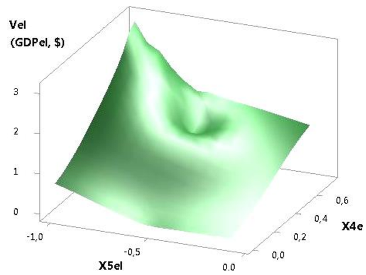


Fig. 3. 3D of Vel (GDPel)

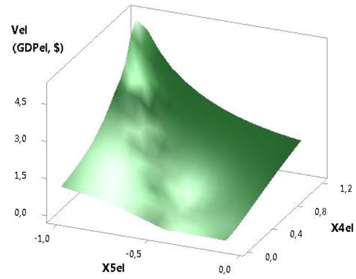


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were X1= 5...-5, X2= -10...-1, X3= -1...-10, X4=0.59...0.73, X5= -0.1...-1 and X1= X2= 5...-5, X3= -1...-10, X4=0.35...1.21, X5= -0.1...-1. As you can see, the plotted 3D Vel graph in Fig. 3 increases by 3.41 times, and in Fig. 4 increases by 13.84 times.

The calculated values for the 3D Vel graph in Figure 5 at variables X1= X2= X3= 5...-5, X4=1.39...1.56, X5= -0.1...-1 increase by 111.84 times. From the following Figure 6, it can be seen that at variables X1= X3= 5...-5, X2= 1.10, X4=0.29.2.87, X5= -0.1...-1, Val values increase 1000 times.

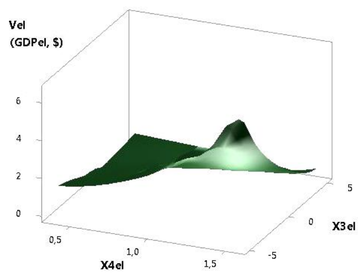


Fig. 5. 3D of Vel (GDPel)

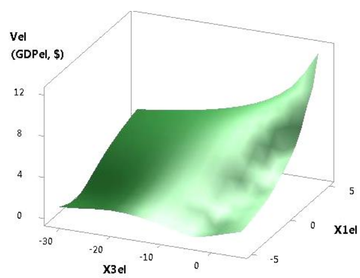


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at X1= 1...10, X2= 5...-5, X3= -1...-10, X4=-0.32...0.18, X5= -0.1...-1 and X1= -10...-1, X2= 5...-5, X3= -1...-10, X4=4.4...0.83, X5= -0.1...-1. Here in Figure 7, the Vel values have a maximum of 0.6 at point 6, and in Figure 8 they increase by 18.91 times.

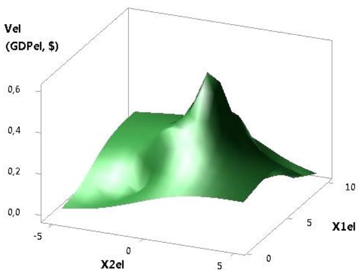


Fig. 7. 3D of Vel (GDPel)

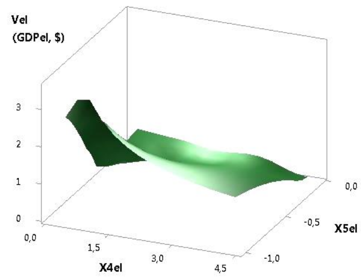


Fig. 8. 3D of Vel (GDPel)

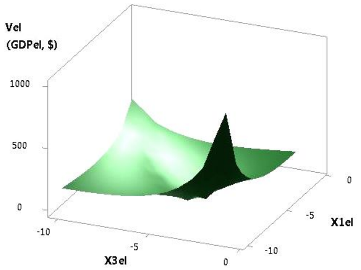


Fig. 9. 3D of Vel (GDPel)

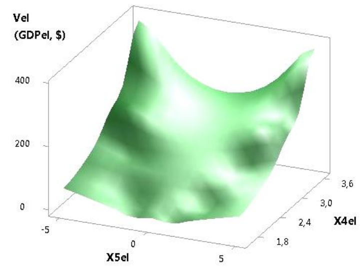


Fig. 10. 3D of Vel (GDPel)

The following figures 9 and 10 were plotted at X1= X2= -10...-1, X3=-1...-10, X4=9.49...3.53, X5= 5...-5 and X1= -10...-1, X2= X3= -1...-10, X4=3.64...3.64, X5= 5...-5. Here in Figure 9, the Vel variable has a minimum of 1.63 at point 6, and in Figure 10 it has a minimum of 1.99 at point 6.

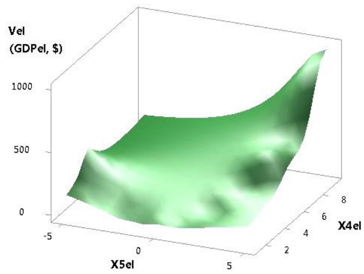


Fig. 11. 3D of Vel (GDPel)

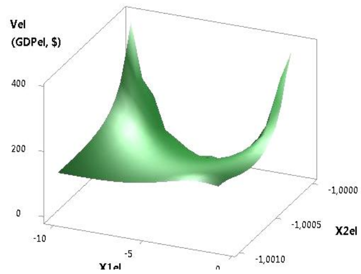


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at X1= X2= -1...-10, X3= -10...-1, X4=3.54...9.48, X5= 5...-5 and X1= -1...-10, X2= -1, X3= -10...-1, X4= 3.54...3.64, X5=5...-5. In these examples in Figure 11, the Vel variable has a minimum of 1.75 at point 5, and in Figure 12 it has a minimum of 0.62 at point 5.

## REFERENCES

Swiss banks issue loans to some customers at a negative rate. We tell you how it became possible // https://news.mail.ru/economics/38091566/?frommail=1

Negative mortgage rates. Now it's real. // http://www.vestifinance.ru / articles/ 52690

The mechanism of negative interest rates on loans // http://kpk-bonus.ru/mexanizm-otricatelnyx-stavok-po-kreditam